

Naman Chetwani

+1 925 895 9674 | naman.chetwani@gmail.com | github.com/coder-2011 | naman.world

PROJECTS

Gemma 4 Inference Megakernel

GitHub repo

- Writing gemma.c, a Gemma 4 megakernel written in CUDA and PTX. Full forward pass is one kernel.

Sinter

GitHub repo

- Building an experimental H100 (sm_90) LLM inference engine that compiles declared model products into standalone CUDA serving repositories.
- Rust owns HTTP, tokenization, continuous batching, and paged or radix KV metadata. C++ owns semantic graphs, resource plans, and source emission. CUDA C++ and ThunderKittens own H100 physical forms.
- An offline build selects a verified exact cover of legal graph regions. Model load validates the declared product against configuration and checkpoint metadata without searching, compiling, tuning, or benchmarking variants.

Writing Kernels

- Writing custom GPU kernels in CUDA, PTX, and ThunderKittens, including flash attention, KDA, LatentMOE, etc.

Snaptokens

GitHub repo

- Built a Hugging Face-compatible Rust BPE tokenizer with exact token-ID parity. It is 2.04x faster than Gigatoken and 46.41x faster than Hugging Face across a 12-tokenizer, 15-host paired-median comparison.
- Hand-written scanners, fused ByteLevel and BPE paths, precomputed flat lookups, contiguous pre-tokenization, and warmed Rayon caches remove generic-regex work and allocation.
- Optional .tkz sidecars cache validated native construction data. Direct .tkz loads are 1.58x faster and artifacts are 24.9% smaller than Snaptokens JSON by geometric mean.

LPT

GitHub repo

- Built a latent planning token path that injects planning tokens every n rollouts, then adds their final-layer hidden states to the next n rollouts.
- Improved perplexity by 4.6% for 0.3% overhead in a matched run; the mechanism should become more useful with scale.

STRATUS UAV: Optimizing Prescribed Burning using UAVs and Predictive Modeling

- Led development of a patent-pending autonomous UAV to mitigate wildfire spread; 3D-printed a VTOL aircraft for autonomous backburning with 15 km range and 1.2 hour endurance.
- Targeted core field constraints: helicopter operations around \$16k/day, custom aircraft and pilots, and pilot liability; designed for 5x lower price and 3x higher range than the closest drone competitor.

SBON (Strategic-Blackline-Optimization-Engine)

GitHub repo

- Built a wildfire prediction model that set state-of-the-art results and created the first ML approach to optimizing prescribed burns.
- Built a tri-branch multimodal architecture across 13 input modalities spanning fire imagery, weather, and geography; ConvNeXt and SSM encoders feed a larger SSM for 4-hour fire-mask prediction.
- Treated the model as an environment, rolled out predictions to 48 hours, and used PPO to optimize burn placement in a niche field with fewer than 20 papers in the last 30 years.

The no-circles project

no-circles.com | GitHub repo

- Built a personalized daily newsletter that learns from each reader's replies and behavior; scaled to triple-digit users within 3 weeks.
- Built the user memory, daily brief, and reply feedback loop to explain unfamiliar fields through topics each reader already cares about.

PHOENIX UAV / Science Fair Results

GitHub repo

- Built PHOENIX UAV for early wildfire detection; placed 2nd in the California State Science Fair, 1st at regionals, and won 5 awards including the Henkel aerospace award and Junior Lemelson-MIT Inventor Award.

EDUCATION

Astra Nova High School

2026-2028

Independent Study / Competitions

- Selected for SPARC '26 (3% acceptance rate); achieved USACO Gold and F=MA qualifying scores in the top 5% nationally.
- Competed in CALICO, Harker Physics Invitational, and Harker Programming Invitational; held a top 30 national placement in MSPF Debate with 2 tournament wins and elimination rounds at Stanford and TOC.